



D Y Patil College of Engineering Akurdi Pune

Department of Robotics and Automation

A-Y 2026-27

SY B.Tech Robotics and Automation

FAB LAB – FIELD SURVEY FOR PROJECT

1. Student Details

Sr. No.	Roll No.	Name	Branch
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2. Survey Details

- **Place/Organization:** Police Station, Ravet
- **Date:** September 05, 2026
- **Person Contacted:** A police officer from Department (Avinash Sir)

3. Problem Identified in Community / Society

- Police personnel are required to perform several repetitive administrative tasks, including maintaining records, documentation, legal paperwork, and courtroom-related paperwork.
- Police personnel are also required to perform duties such as night patrolling, VIP management, security arrangements, and handling various public events. This can result in limited personnel availability for other routine activities.
- Physical evidence collected during investigations requires proper storage, handling, movement, and record maintenance.
- Evidence may be transferred between different authorized personnel or locations, such as the evidence storage area, investigating officers, forensic laboratories, and courts.

- Maintaining a continuous and accurate manual record of evidence movement can be time-consuming and may increase the possibility of missing or incorrectly recording information.
- Therefore, there is a need for a system that can assist in **monitoring evidence movement and maintaining a systematic digital log**.

4. Observations & Data Collected

- Evidence requires continuous management, tracking, and record maintenance throughout its custody period.
- Evidence can be handled by multiple authorized personnel during an investigation.
- Important information such as **evidence identification, person handling the evidence, time, location, and movement status** needs to be recorded.
- Manual recording of every evidence movement can involve repetitive paperwork and requires personnel time.
- A digital identification and logging system could reduce repetitive record-keeping work.
- RFID technology can be used to provide each evidence package with a unique identification.
- A centralized or local digital log can maintain the movement history of each evidence item.
- Additional monitoring features such as **tamper detection and unauthorized movement alerts** can improve the monitoring capability of the system.
- The system should support existing police and legal procedures rather than replace the official chain-of-custody process.

5. Possible Solution (using Robotics/Automation/FAB Lab tools)

The proposed solution is **EviLog – Smart Evidence Monitoring and Chain-of-Custody System**.

EviLog will assign a unique RFID tag to each evidence package. Whenever an evidence item is stored, issued, transferred, received, or returned, the RFID tag can be scanned using an RFID reader. The system will record the relevant information, such as:

- Evidence ID
- Date and time
- Person/Officer handling the evidence
- Current location
- Type of transaction
- Evidence status
- Condition of the evidence/package

A tamper-detection mechanism can also be incorporated into the evidence container. If the container is opened or an unexpected movement is detected, the system can generate an alert for verification.

The basic working flow will be:

Evidence Registration → RFID Identification → Storage → Issue/Transfer → Scan at Destination → Digital Logging → Return/Storage

This system will provide a traceable digital history of evidence movement and reduce dependence on repetitive manual logging.

6. Key Learning / Outcome

- Identified a real-world problem related to **evidence management and chain-of-custody monitoring** through field interaction with police personnel.
- Understood the importance of maintaining accurate records during the movement and handling of physical evidence.
- Learned how repetitive manual processes can be identified as potential areas for automation.
- Understood the importance of tracking the **identity, location, handler, time, and status** of an object throughout its movement.
- Learned that an automated system should complement existing operational and legal procedures rather than replace authorized decision-making.
- The survey helped establish the requirements for developing a **low-cost prototype of EviLog** as a practical Robotics and Automation project.

7. Signatures

- **Student:** _____
- **Faculty In-Charge:** _____